Objective part

1. Converges process in Regula Falsi Method is ______ than bisection method.

a) ~~Ra~~ Faster   b) Slower   c) None

2. In Iterative method initial solution of given system is ______

a) Many others   b) Exact   c) At least one

3. ______ method is also ... the method of tangent.

a) Newton's Raphsom b) Iterative method d) Jacobi's Method

4- It is that quantity of error which is present in the statement of the problem itself, before its solution

a) Inherent Error b) Local truncation error

c) Local round-off error d) Typing Error

5- A Sufficient condition for converges of the Iterative Solution to the exact Solution:

a) $|a_{ij}| < \sum_{\substack{j=1 \\ j \neq 1}}^{n} |a_{ij}|$ b) $|a_{ij}| \leq \sum_{\substack{j=1 \\ j \neq 1}}^{n} |a_{ij}|$

c) $|a_{ij}| > \sum_{\substack{j=1 \\ j \neq 1}}^{n} |a_{ij}|, i=1,2,3$ d) $|a_{ij}| \geq \sum_{\substack{j=1 \\ j \neq 1}}^{n} |a_{ij}|$

6- Jacobi's Method is same as th

a) Guass - Seidel b) Relaxation method

c) Guass - Elimination d) Bisection Method

7. Eigen value is

a) Diagond b) scalar

c) vector d) Scalar and vector

The Square matr

is ______

a) Non-Singular ✓ b) Singular

c) Diagonal d) None

9. Near a simple root Muller's method converges ___ than the secant method.

a) faster

b) Slower

10. Which method is required a derivate of a Solution

a) Regula Falsi Method b) Muller method

c) Newton Raphson method d) Bisection method

11. Hexadecimal of number system is .

a) 2 b) 8

c) 10 d) ✓ 16

12. Every equation of the form $f(x) = 0$ has

a) $>$ zero b) ✓ At least one

c) one d) None of these

13. The number of positive roots of an alegbraic equation $f(x)$ = ____

a) 1 b) -1

c) 0 d) None

14. The Initial Interval (x_1, x_2) in which the root of the equation lies should be ______

a) Small b) Large

c) Equal d) None

15. In the Newton-Raphson Method, the root is ______

a) bracketed

b) Not bracketed

16. The secant method converges ___ than Newton's quadratic

a) Faster

b) Slower

17. $2x+3y-z=5$

$4x+4y-3z=3$

$-2x+3y-z=1$ is a ______

a) ✓ Unique b) No Solution

c) Infinit many d) None of these

18. Relaxation Method is a/an

a) Direct Method b) Iterative Method

19. Symbol used for forward difference is

a) ✓ Δ b) ∇

c) δ d) μ

20- Secant's method required

a) One-point b) Two-point

c) Three-point d) None-

21- The method of computing the value of y, for a given value of x lying outside the table of value of x.

a) Iterpolation b) extrapolation

22. The matrix is real and symmetri
The largest - diagonal Element

a) $A^{-1}A = I$ b) $A^{-1}A = S$

c) $A^{-1}A = D$ d) $A^{-1}A = [0]$

23- Orthogonal matric, then the diagonal Element

a) $S^{-1}DS = I$ b) $SDS^{-1} = [0]$

24- An $n \times n$ matrix $[A]$ is said to be orthogonal if

a) $A^T A = I$ b) $[A]^{-1}[A] = I$

25- Which of the following is the meaning of partial pivoting while employing the row transformation?

a) Making largest as pivot b) Making smalles element as pivot

c) Making any element as pivot d) Making zero element as pivot

Short Questions

Question #1:- $f(0) = -4$, $f(2) = 6$, then find the next appromation value the function using secant method.

$$f(0) = -4, \quad f(2) = 6, \quad f(x_0) = -4, \quad f(x_1) = 6$$

The other values $x_0 = 0$, $x_1 = 2$

In secant method,

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{}$$

In secant method,

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{0(+6) - 2(-4)}{6 - (-4)} = \frac{0+8}{6+4} = \frac{8}{10}$$

$$x_2 = 0.8$$ Answer

Question # 2:- Find the value of θ from the following by Jacobi's Method

$$A = \begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix}$$

The given matrix is real and symmetric. The

$$a_{12} = a_{21} = \frac{1}{2}$$

Now we complute

$$\tan 2\theta = \frac{2a_{ij}}{a_{ii} - a_{jj}} = \frac{2a_{12}}{a_{11} - a_{22}}$$

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$$= \frac{\cancel{2}\left(\frac{1}{\cancel{2}}\right)}{1 - \frac{1}{3}} = \frac{1}{\frac{3-1}{3}}$$

$$= \frac{1}{2/3} = \frac{3}{2}$$

$$= \frac{1}{2/3} = \frac{3}{2}$$

$$\tan 2\theta = \frac{3}{2}$$

$$2\theta = \tan^{-1}\left(\frac{3}{2}\right)$$

$$\theta = \frac{\tan^{-1}\left(\frac{3}{2}\right)}{2} = 28.15$$

Q 3: Find

20.155

Q#3. Find an Interval in which root of the equation $x^3-9x+1=0$ lies.

$$f(x) = x^3 - 9x + 1 = 0$$

Now $f(2) = (2)^3 - 9(2) + 1 = 8 - 18 + 1 = -9$

$f(4) = (4)^3 - 9(4) + 1 = 64 - 36 + 1 = 29$

Both have opposite sign $(-9)(29)$ ...

$f(2)\,f(4) <$ ...

of the equation $x^3 - 9x + 1 = 0$ lies.

$f(x) = x^3 - 9x + 1 = 0$

Now $f(2) = (2)^3 - 9(2) + 1 = 8 - 18 + 1 = -9$

$f(4) = (4)^3 - 9(4) + 1 = 64 - 36 + 1 = 29$

Both have opposite sign $(-9)(29)$, So $f(2)\,f(4) < 0$ So roots lies between 2 and 4.

Question # 1:- Solve the following system of equation by Jacobi's Iteration method to three decimal places upto one Iteration

$$83x + 11y - 4z = 95$$

$$7x + 52y + 13z = 104$$

$$3x + 8y + 29z = 71$$

Taking Initial $(0\ \ 0\ \ 0)^T$

$$x = \frac{95}{83} - \frac{11}{83}y + \frac{4}{83}z$$

$$y = \frac{104}{52} - \frac{7}{52}x - \frac{13}{52}z$$

$$z = \frac{71}{29} - \frac{3}{29}x - \frac{8}{29}y$$

1st Iteration $(0 \quad 0 \quad 0)^T$

$$x = \frac{95}{83} - \frac{11}{83}(0) + \frac{4}{83}(0) = 1.1446$$

$$y = \frac{104}{52} - \frac{7}{52}(0) - \frac{13}{52}(0) = 2.000$$

$$z = \frac{71}{29} - \frac{3}{29}(0) - \frac{8}{29}(0) = 2.448$$

$$\begin{pmatrix} x_{(1)} \\ y_{(1)} \\ z_{(1)} \end{pmatrix} = \begin{pmatrix} 1.144 \\ 2.000 \\ 2.448 \end{pmatrix}$$

$$x^{(2)} = \frac{95}{83} - \frac{11}{83}(2.000) + \frac{4}{83}(2.448) = 0.9978$$

$$y^{(2)} = \frac{104}{52} - \frac{7}{52}(1.144) - \frac{13}{52}(2.4483) = 1.233$$

$$z^{(2)} = \frac{71}{29} - \frac{3}{29}(1.144) - \frac{8}{29}(2.00) = 1.742$$

Question #2:- Approximate the root $x^3-2x-5=0$ between $(2,3)$ by using Regula-False Method upto two Iteration

$f(x) = x^3 - 2x - 5$

$f(2) = (2)^3 - 2(2) - 5 = 8 - 4 - 5 = -1$

$f(3) = (3)^3 - 2(3) - 5 = 27 - 6 - 5 = 16$

Since $f(2)$ $f(3)$ are of opposite sign, the root

Taking $x_1 = 2$, $x_2 = 3$ and using Regula-Falsi method, the first approximation is given by

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$= 3 - \frac{3-2}{16-1}(16)$$

$$= 3 - \frac{1}{15}(16) \Rightarrow 3 - 1.066$$

$$x_3 = 1.933$$

$$f(x_3) = x^3 - 2x - 5$$

$$= (1.933)^3 - 2(1.933) - 5$$

$$= 13.97 - 3.866$$

$$= 5.104$$

Now, Since $f(x_1)$ and $f(x_3)$ are of opposite sign, the second approximation is

$$x_4 = x_3 - \frac{x_3 - x_1}{f(x_3) - f(x_1)} f(x_3)$$

$$= 1.933 - \frac{1.933 - 2}{5 - (-1)} \times 5.104$$

$$= 1.933 - \frac{-0.067}{6} \times 5.104$$

$$= 1.933 + 0.0569$$

$x_4 = 1.989$

$f(x_4) = x^3 - 2x - 5$

$= (1.989)^3 - 2(1.989) - 5$

$= 15.682 - 3.978 - 5$

$f(x_4) = 6.704$

Question # 3:- Find the dominant eigen value and the eigenvector of the matrix

$$\begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 9 & 15 \end{bmatrix}$$ by Power method

the matrix

$$\begin{bmatrix} 8 & 1 & 2 \\ 0 & 10 & -1 \\ 6 & 2 & 15 \end{bmatrix}$$ by power method

vector $V^0 = [1 \quad 1 \quad 1]^T$ as the initial vector

(Answer Required two Iteration)

We choose an initial vector $v^{(0)}$ as $(1\ 1\ 1)^T$

$$u^{(1)} = [A]v^{(0)} = \begin{bmatrix} 8 & 1 & 2 \\ 0 & 16 & -1 \\ 6 & 2 & 15 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 8+1+2 \\ 0+16-1 \\ 6+2+15 \end{bmatrix} = \begin{bmatrix} 11 \\ 15 \\ 23 \end{bmatrix}$$

$$u^{(1)} = 23 \begin{bmatrix} 11/23 \\ 15/23 \end{bmatrix}$$

vector

$$u^{(1)} = 23 \begin{bmatrix} 0.478 \\ 0.652 \\ 1.00 \end{bmatrix}$$

The Second Iteration

$$u^{(2)} = [A]\, v^{(1)}$$

$$= \begin{bmatrix} 8 & 1 & 2 \\ 0 & 16 & -1 \\ 6 & 2 & 15 \end{bmatrix} \begin{bmatrix} 0.478 \\ 0.652 \\ 1.00 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 1 & 2 \\ 0 & 16 & -1 \\ 6 & 2 & 15 \end{bmatrix} \begin{bmatrix} 0.478 \\ 0.652 \\ 1.00 \end{bmatrix}$$

$$= \begin{bmatrix} 3.824 + 0.652 + 2 \\ 0 + 10.432 - 1 \\ 2.868 + 1.304 + 15 \end{bmatrix}$$

$$= \begin{bmatrix} 6.476 \\ 9.432 \\ 19.112 \end{bmatrix}$$

Now, Normalize

$$= 19.172 \begin{bmatrix} 6.476/19.172 \\ 9.432/19.72 \\ 19.172/19.72 \end{bmatrix}$$

$$= 19.172 \begin{vmatrix} 0.491 \\ 1.00 \end{vmatrix}$$

Question #3:- Constract the Forward difference table from the following data.

x :	5	10	15	20	25	30
y :	9962	9848	9659	9397	9063	8660

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
5	9962	9848-9962=-114	-189-(-114)=-75	
10	9848	9659-9848=-189	-73	-73+75
15	9659	9397-9659=-262	-72	1
20	9397	9063-9397=-334	-69	3
25	9063	8660-9063=-403		

Question #3:- Solving the following system of linear equation by Gauss-seidel Iteration method up to one iteration.

$$10x + 2y + z = 13$$

$$2x + 10y + z = 13$$

$$2x + y + 10z = 13 \qquad (0, 0, 0)^T$$

Taking the initial solution as

Other Method - Iteration-I

$$x^{(1)} = \frac{13}{10} - \frac{2}{10}(0) - \frac{1}{10}(0) = 1.3$$

$$y^{(1)} = \frac{13}{10} - \frac{2}{10}(1.3) - \frac{1}{10}(0) = 1.04$$

$$z^{(1)} = \frac{13}{10} - \frac{2}{10}(1.3) - \frac{1}{10}(1.04) = 0.936$$

Question # 1.

Convert the following matrix into diagonal form.

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & -3 & -9 \\ 0 & 0 & -5 \end{bmatrix}$$

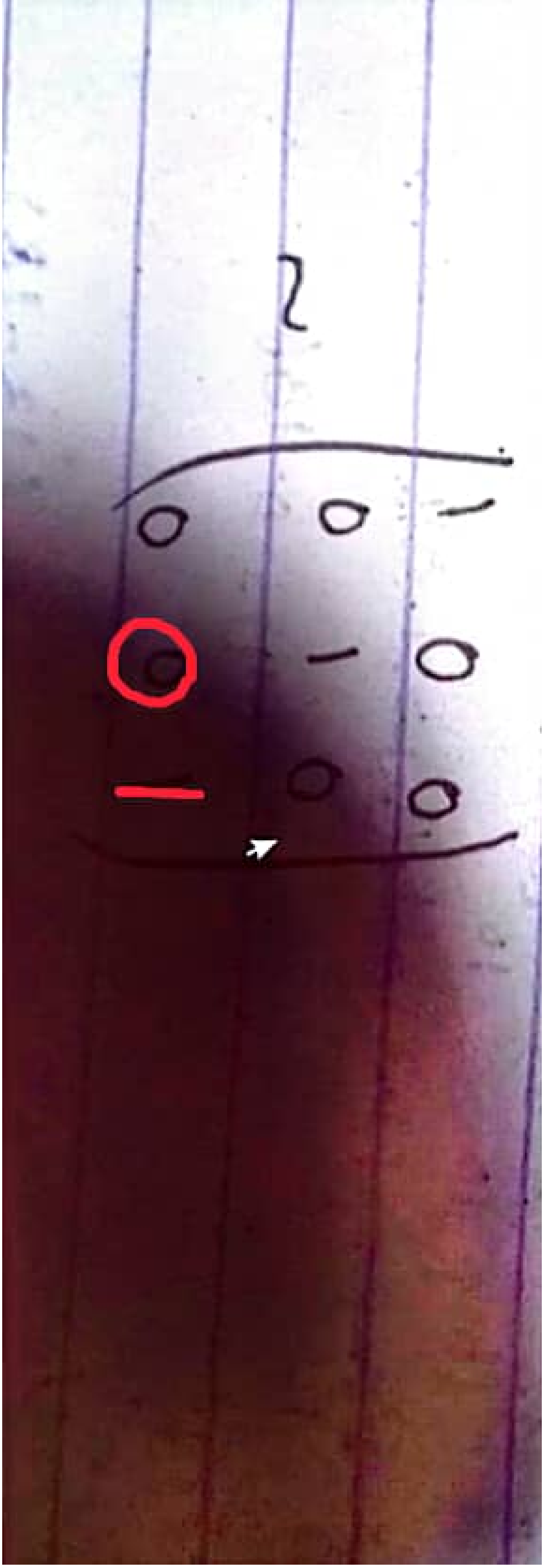
$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & -3 & -9 \\ 0 & 0 & -5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & -9 \\ 0 & 0 & -5 \end{bmatrix} (R_1 + R_3)$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & -5 \end{bmatrix} -\frac{1}{3}R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \frac{1}{5} R_3$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_2 - 3R_1$$



Q#2: Find the largest eigen value and the corresponding eigen vector of the matrix $\begin{pmatrix} 1 & 2 & -5 \\ -5 & 2 & 8 \\ 1 & 2 & 4 \end{pmatrix}$. By power method at the end of two Iteration taking $\begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix}$ as the initial vector.

$$\begin{pmatrix} 1 & 2 & -5 \\ -5 & 2 & 8 \\ 1 & 2 & 4 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} -1+4-20 \\ 5+4+32 \\ -1+4+16 \end{pmatrix} = \begin{bmatrix} -17 \\ 41 \\ 19 \end{bmatrix}$$

Normalizing the result, we get

$$41 \begin{bmatrix} -17/41 \\ 1 \\ 19/41 \end{bmatrix} = 41 \begin{bmatrix} -0.41463 \\ 1 \\ 0.463 \end{bmatrix}$$

Again

$$\begin{bmatrix} 1 & 2 & -5 \\ -5 & 2 & 8 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} -0.41463 \\ 1 \\ 0.463 \end{bmatrix}$$

$$\begin{bmatrix} -0.41463 + 2 - 2.315 \\ 2.073 + 2 + 3.704 \\ -0.41463 + 2 + 1.852 \end{bmatrix} = \begin{bmatrix} -0.7316 \\ 7.780 \\ 3.439 \end{bmatrix}$$

$$7.780 \begin{bmatrix} -0.731/7.780 \\ 1 \\ 3.439/7.78 \end{bmatrix}$$

Question # 4:- Solve the following System of equation using Gaussian Elimination Method.

$$x_1 + 5x_2 = 7$$
$$-2x_1 - 7x_2 = -5$$

$$x_1 + 5x_2 = 7 \quad — (1)$$
$$-2x_1 - 7x_2 = -5 \quad — (2)$$

... equation (1) and then Add ~~Subtract~~

$$\cancel{2x_1} + 10x_2 = 14$$

$$\underline{-\cancel{2x_1} - 7x_2 = -5}$$

$$3x_2 = 9$$

$$x_2 = \frac{9}{3}$$

$$\boxed{x_2 = 3}$$

put in equation(1)

$$x_1 + 5x_2 = 7$$

$$x_1 + 5(3) = 7$$

$$x_1 = 7 - 15 = -8$$

$$x_1 + 5x_2 = 7$$

$$x_1 + 5(3) = 7$$

$$x_1 = 7 - 15 = -8$$

$$\boxed{x_1 = -8}$$